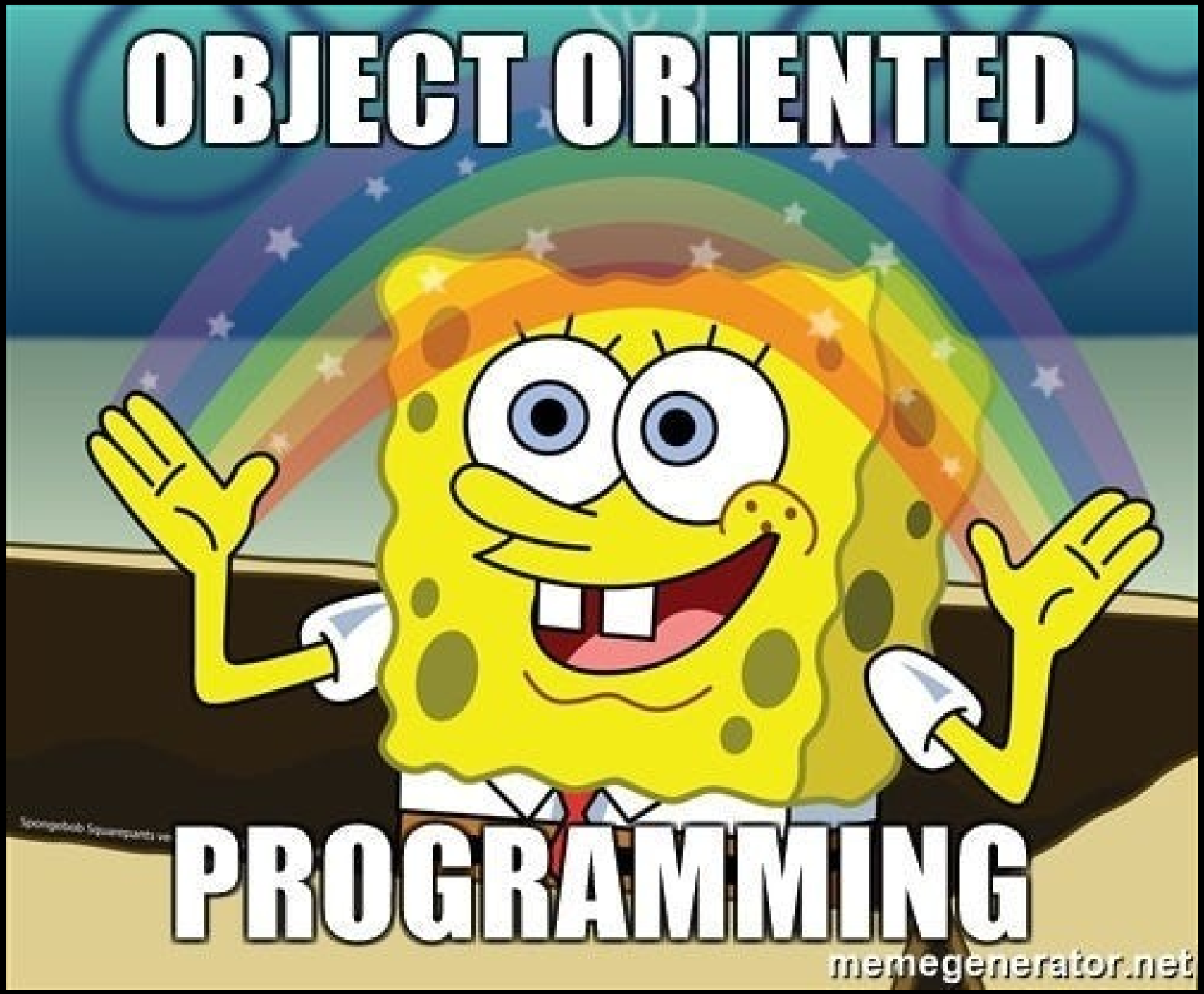
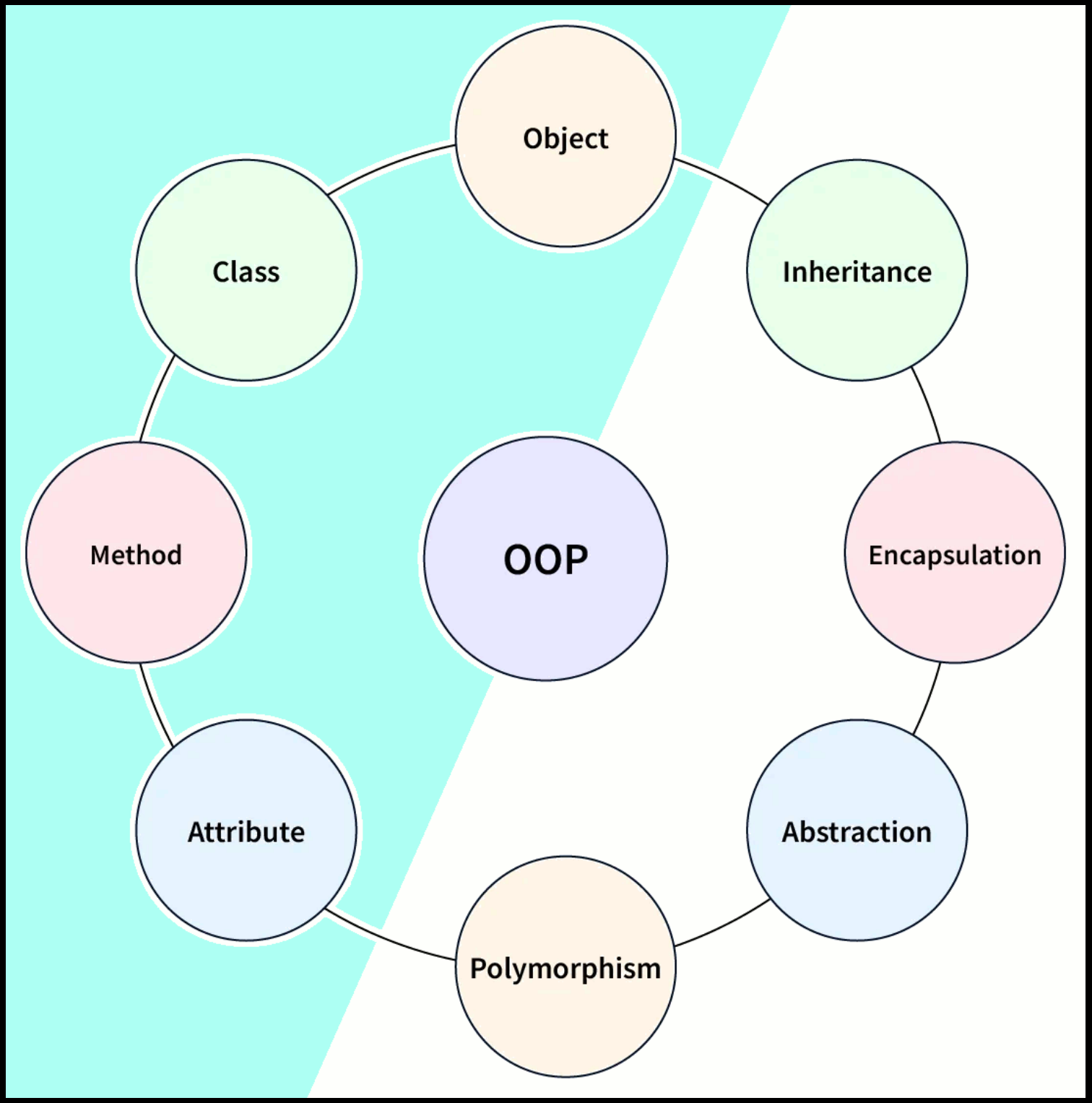
ANÁLISIS DE SISTEMAS

Modelado de Objetos

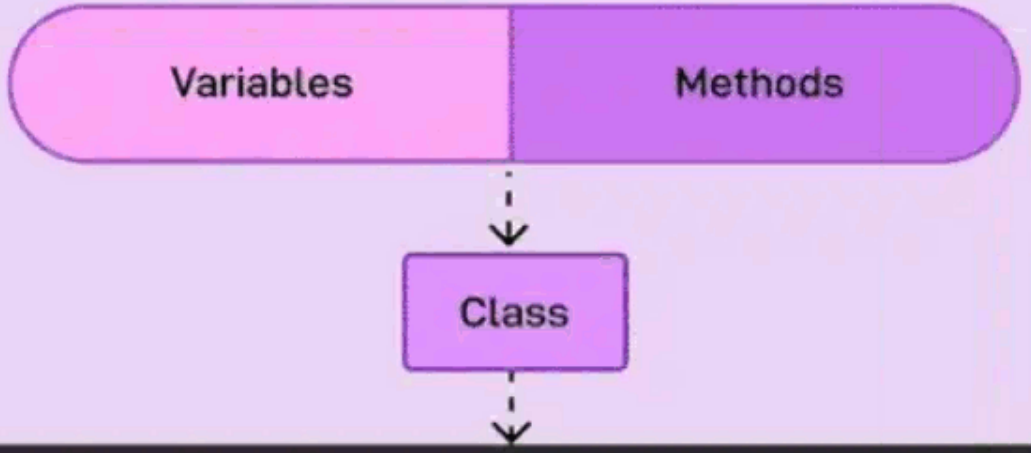

OBJECT ORIENTED
PROGRAMMING
memegenerator.net


Object
Class
Inheritance
Method
OOP
Encapsulation
Attribute
Abstraction
Polymorphism

Encapsulamiento

Encapsulation

- Process of wrapping code and data together into a single unit. It means each object in the code should control its own state




```java
public class Car {
    // private variable
    private String name;

    // getter method for name
    public String getName() {
        return name;
    }

    // setter method for name
    public void setName(String name) {
        this.name = name;
    }
}
```

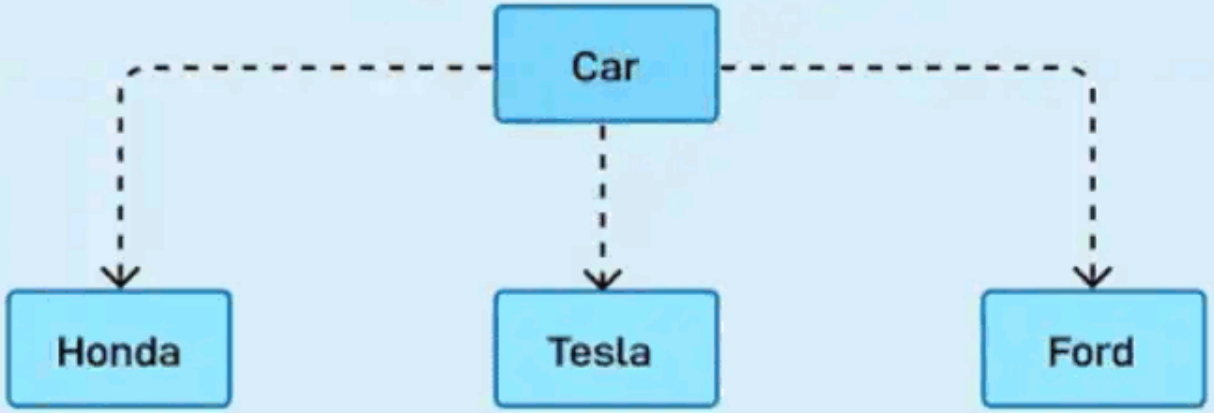
Variables

Methods

Abstracción

Abstraction

- Process of hiding implementation details and exposing only the functionality to the user.




```java
public abstract class Car {
    public abstract void stop();
}

// Concrete class
public class Honda extends Car {
    @Override public void stop()
    {
        System.out.println("Honda::Stop");
        System.out.println(
            "Mechanism to stop the car using break");
    }
}
```

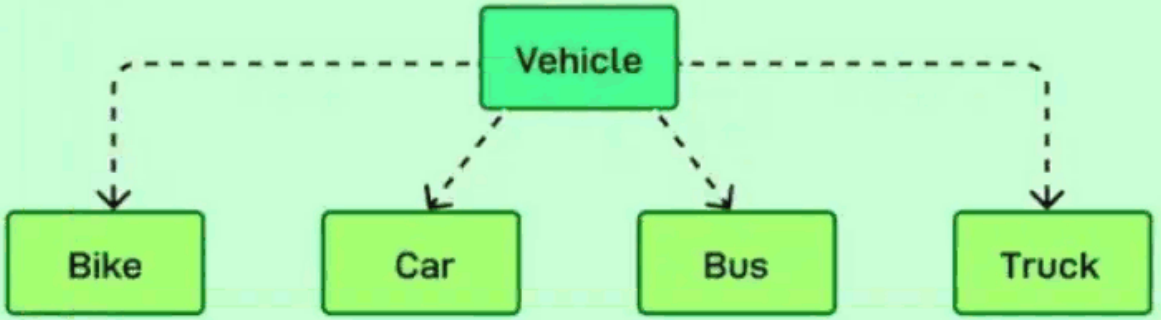
Abstract Class and Method

Providing Implementation

Herencia

Inheritance

- Process of one class inheriting properties and methods from another class. Used to represent the IS-A relationship between objects



Base Class

```java
class Vehicle {
    protected String brand;

    // Constructor
    public Vehicle(String brand)
{
        this.brand = brand;
    }

    // Method to be overridden
    public String describe() {
        return "This is a
vehicle of brand: " + brand;
    }
}
```

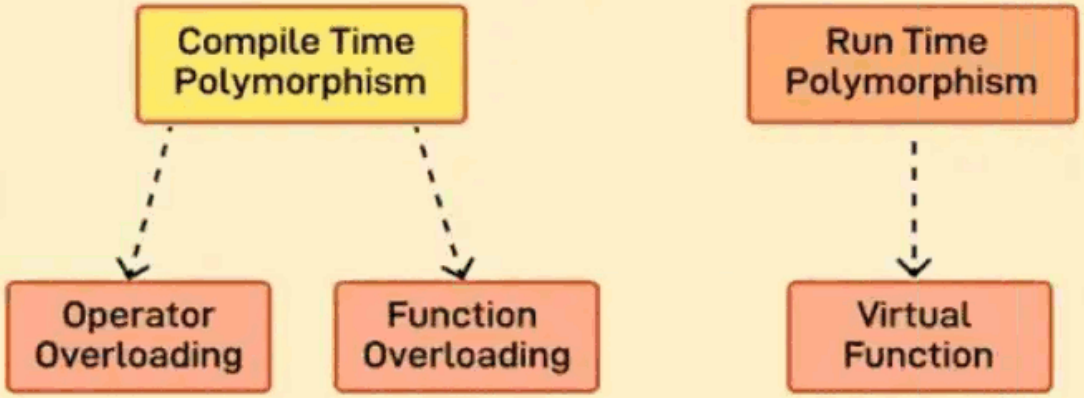
Derived Class

```java
class Car extends Vehicle {
    private int doors;

    // Constructor
    public Car(String brand, int
doors) {
        super(brand);
        this.doors = doors;
    }

    // Override the describe method
    @Override
    public String describe() {
        return "This is a car of
brand: " + brand
            + " with " + doors +
" doors.";
    }
}
```

Polimorfismo

Polymorphism

- Ability to perform many things in many ways. When two types share an inheritance chain, they can be used interchangeably with no errors. This also means using parents like their children



Definition

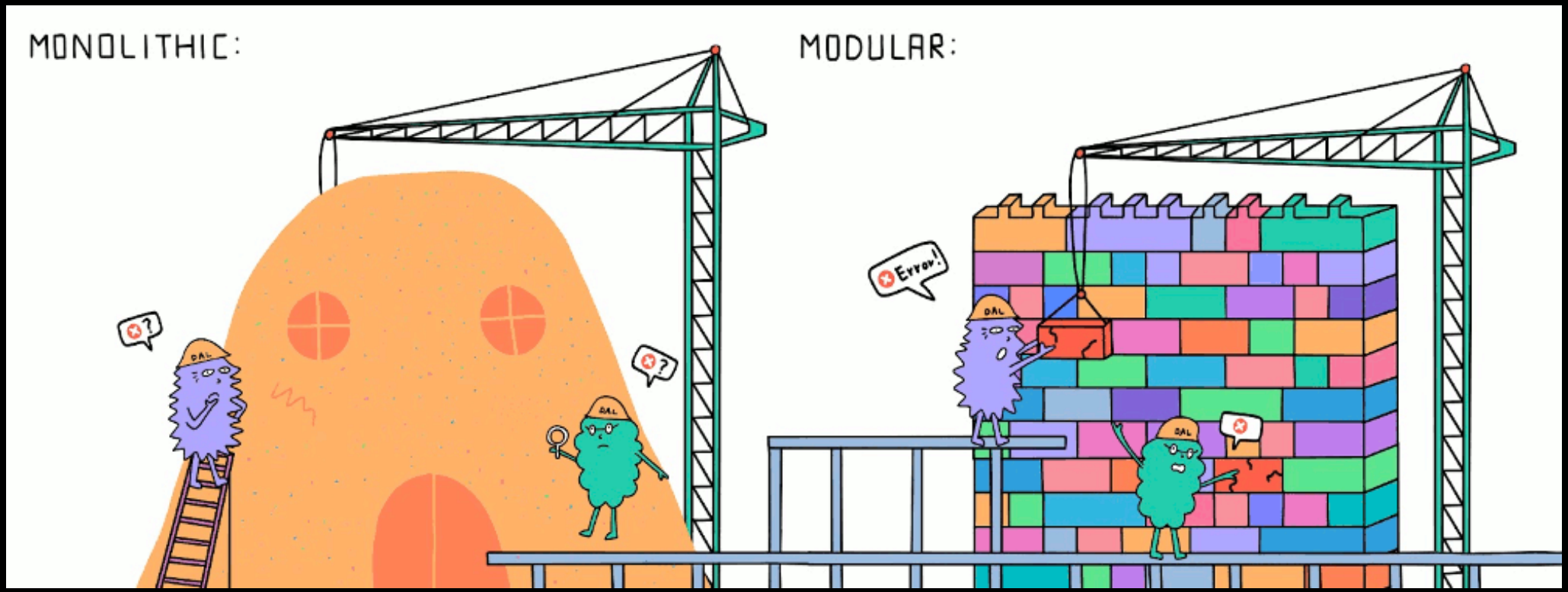
```
class Vehicle {
    public void start() {
        System.out.println("The
vehicle starts.");
    }
}

// Derived class 1
class Car extends Vehicle {
    @Override
    public void start() {
        System.out.println("The
car starts with a key.");
    }
}
```

Usage

```
Vehicle myVehicle;  // Reference
of type

 // Assign Car object to the
Vehicle reference
myVehicle = new Car();

myVehicle.start();
```

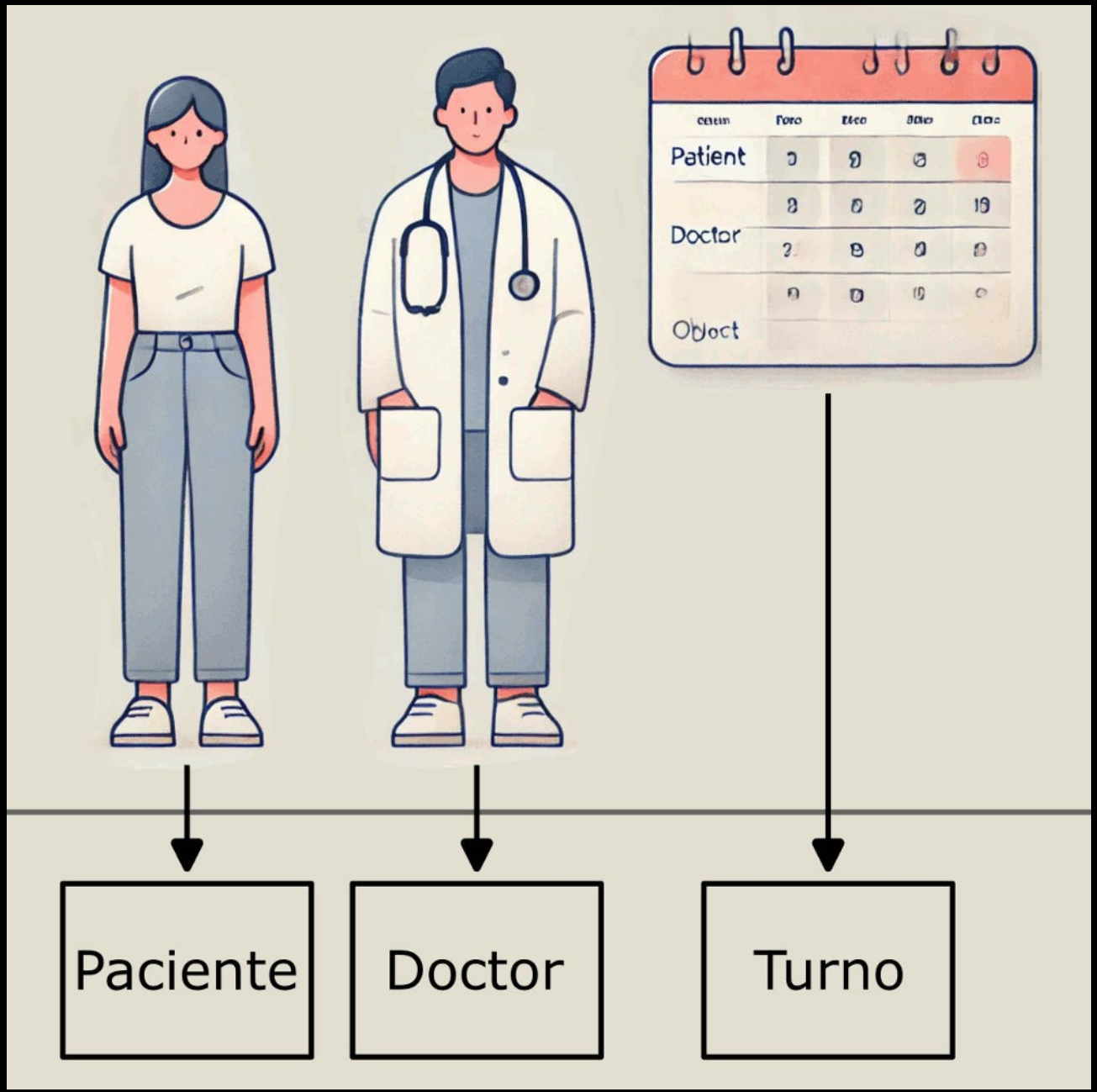
Análisis Orientado a Objetos

- **Modular**


MONOLITHIC:
MODULAR:
Error!
DAL

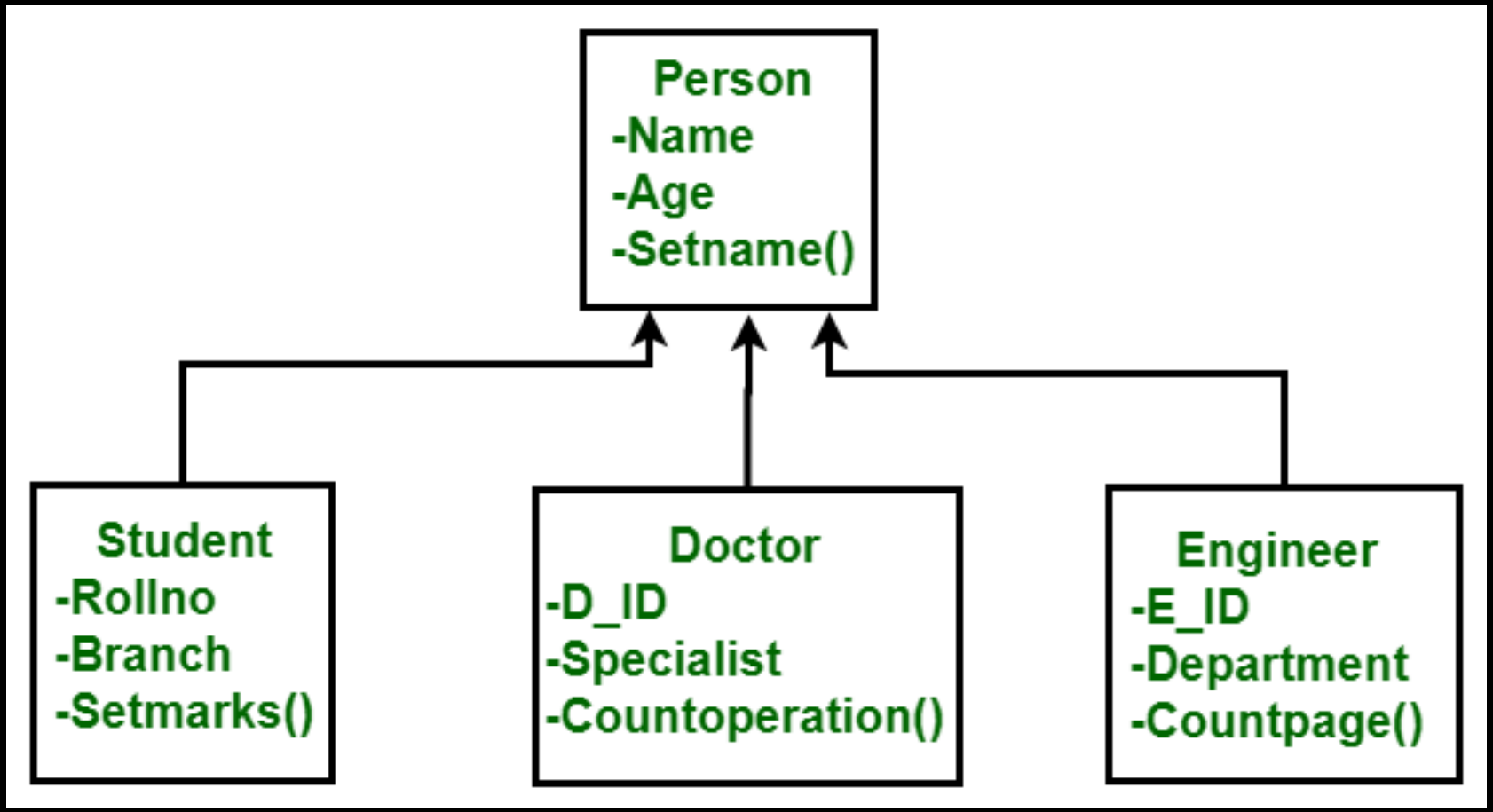
Análisis Orientado a Objetos

- Modular
- **Objetos**


Patient
Doctor
Paciente
Doctor
Turno

Análisis Orientado a Objetos

- Modular
- Objetos
- **Modelo**


Person
-Name
-Age
-Setname()
Student
-Rollno
-Branch
-Setmarks()
Doctor
-D_ID
-Specialist
-Countoperation()
Engineer
-E_ID
-Department
-Countpage()

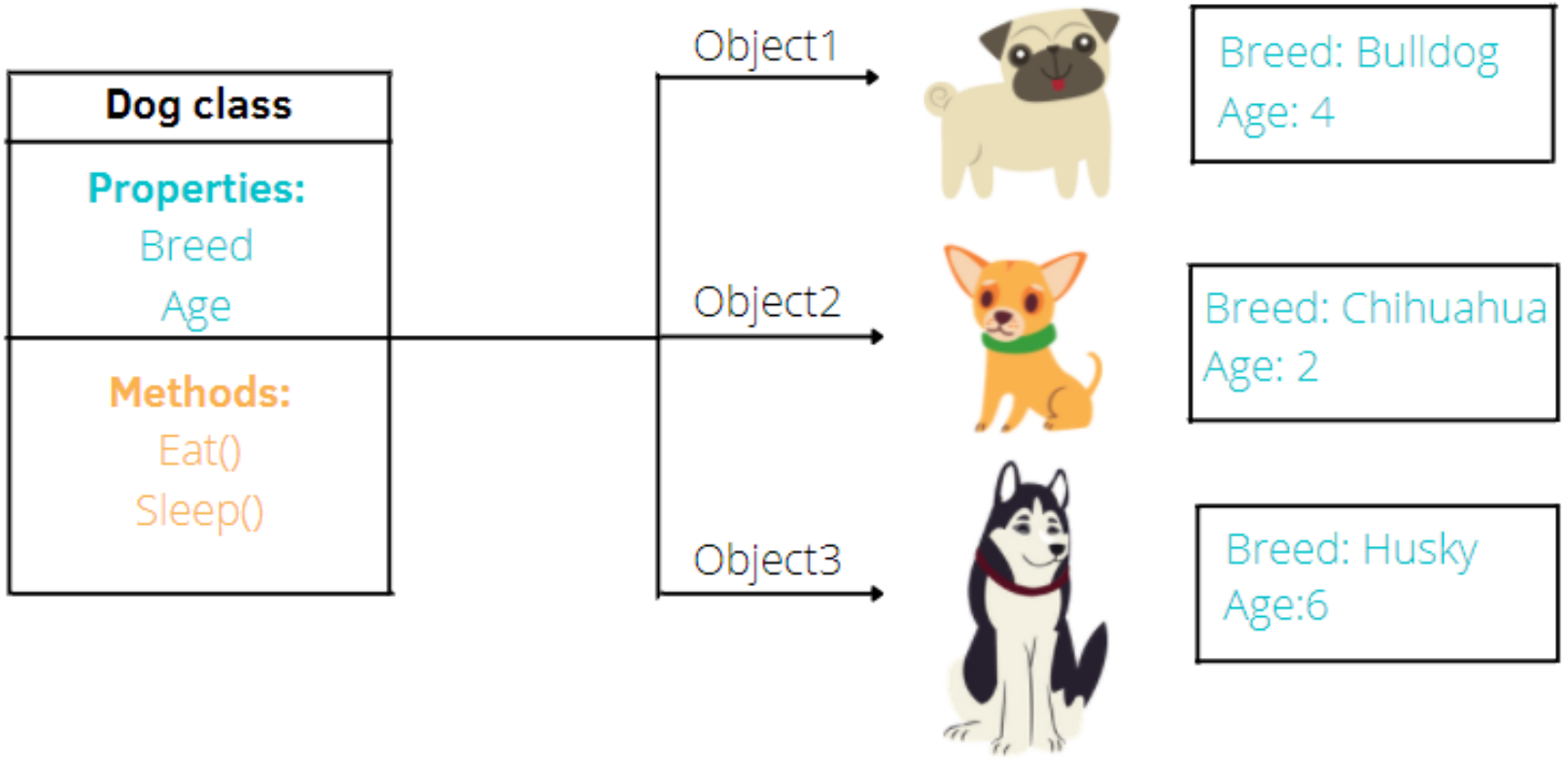
Objetos
(sustantivos)

Atributos
(adjetivos)

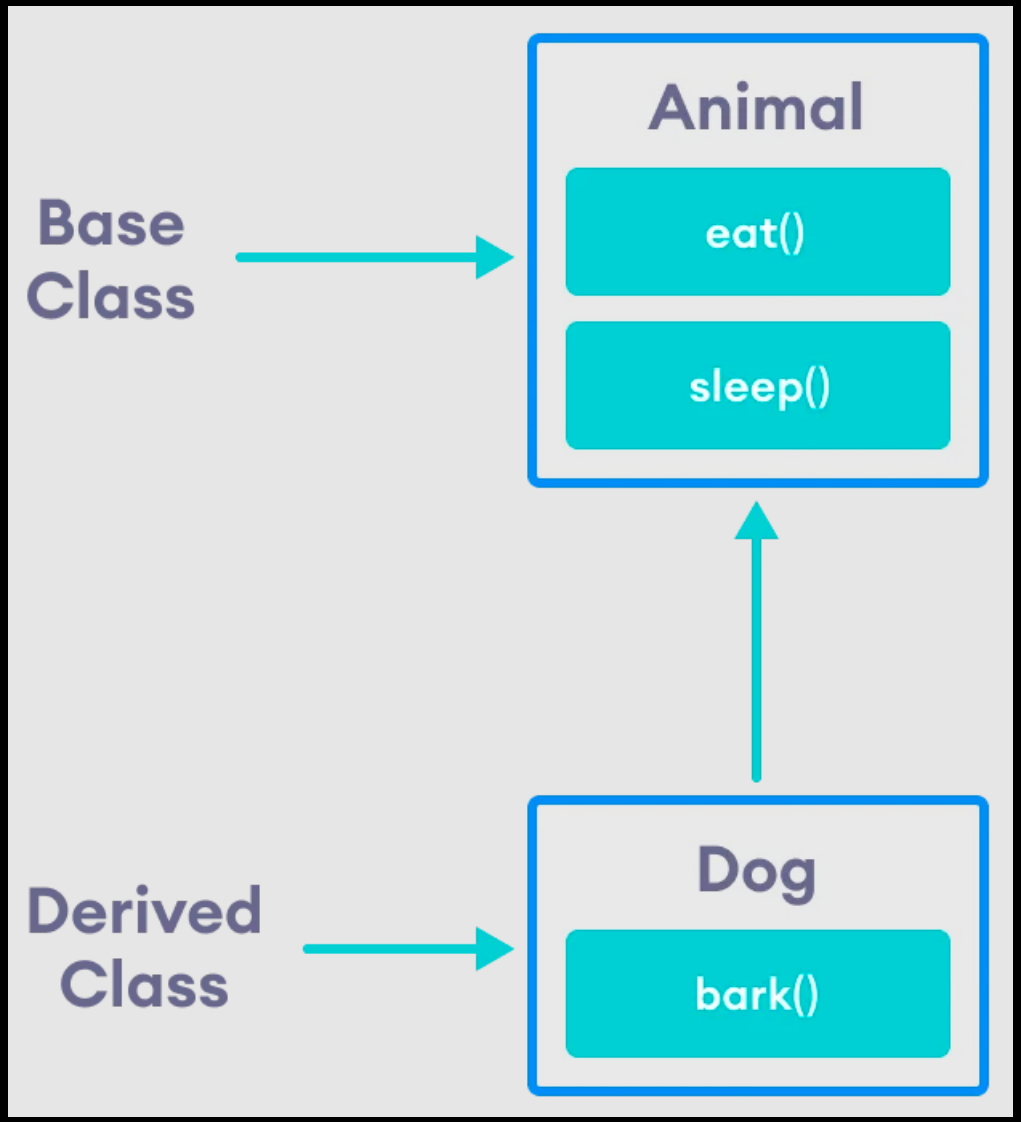
Métodos
(verbos)

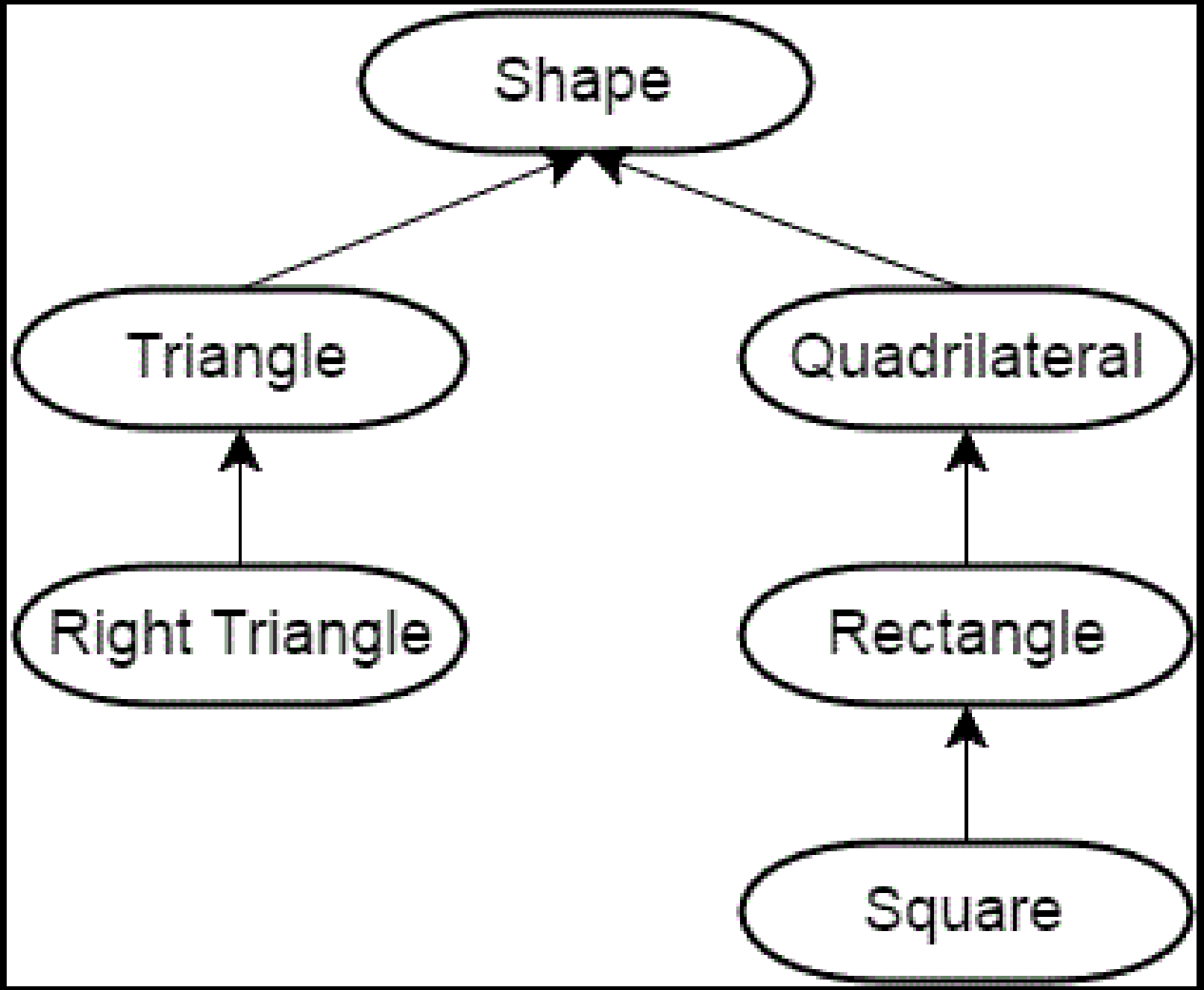
Mensajes

Clases


Dog class
Properties:
Breed
Age
Methods:
Eat()
Sleep()
Object1
Object2
Object3
Breed: Bulldog
Age: 4
Breed: Chihuahua
Age: 2
Breed: Husky
Age:6

Relaciones entre Objetos y Clases

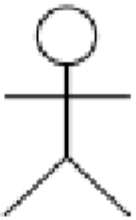
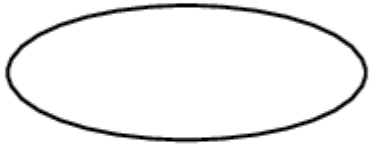
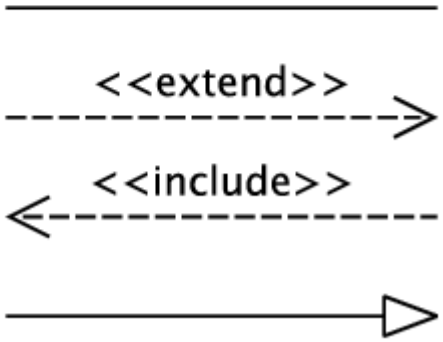



Shape
Triangle
Quadrilateral
Right Triangle
Rectangle
Square

Lenguaje Unificado de Modelado

Lenguaje Unificado de Modelado

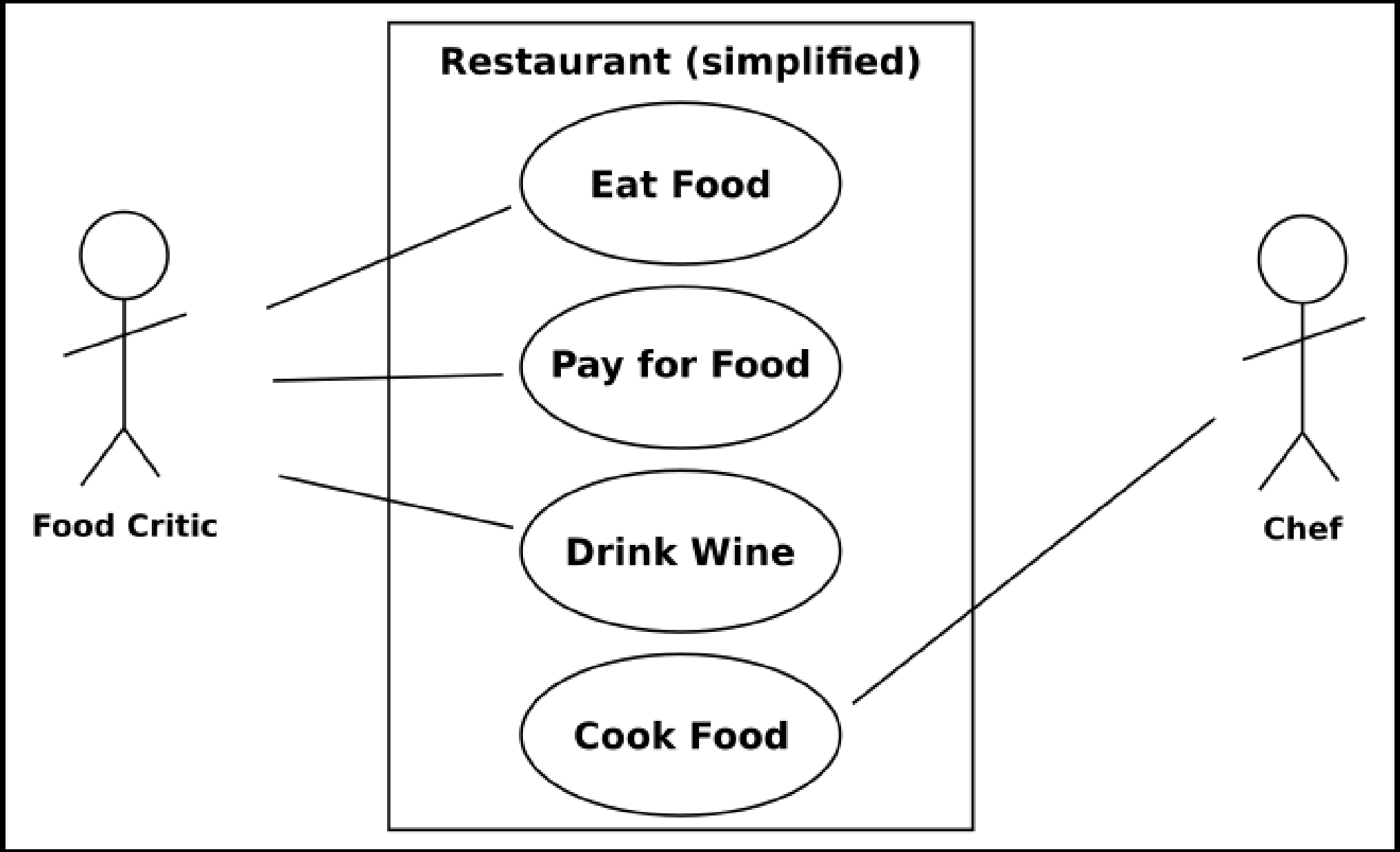
- **Modelado de Casos de Uso**

Symbol	Reference Name
	Actor
	Use case
	Relationship

Use Case description – Retrieve New Password		
High-level description		
	A NetCom Personal Customer has lost the password and requests a new one. Can be used to change the password. The function is used by other NetCom web-portal.	
Actors		
	- NetCom Personal Customer starts the use case, called user in the use case. - Profile server, the data storage of profile data. - NetCom web-portal, the originating system for the request, e.g. MinSide, Mother or mms.netcom.no. - SMSC (SMS center)	
Pre-conditions		
	P1.	The user has accessed this system from another web-based self-service and information systems at NetCom.
Post-conditions		
	Undefined	
Flow of events		
	1.	The use-case starts when the user requests to retrieve a new password. If msisdn is not given in the URL, the user is asked for msisdn.
	2.	The system gets the secret question and answer from the profile

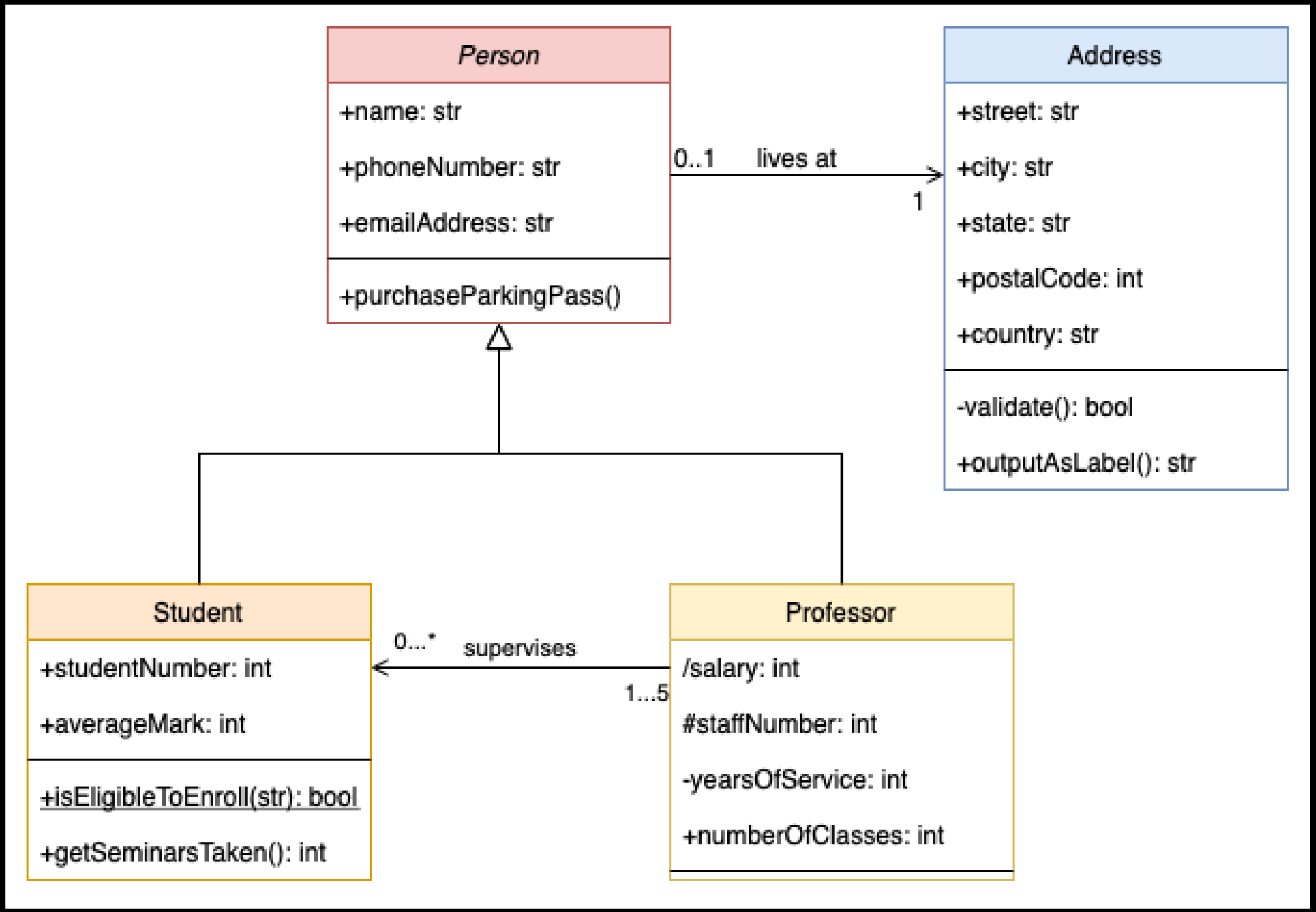
Lenguaje Unificado de Modelado

- Modelado de Casos de Uso
- **Diagramas de Casos de Uso**


Restaurant (simplified)
Eat Food
Pay for Food
Drink Wine
Cook Food
Food Critic
Chef

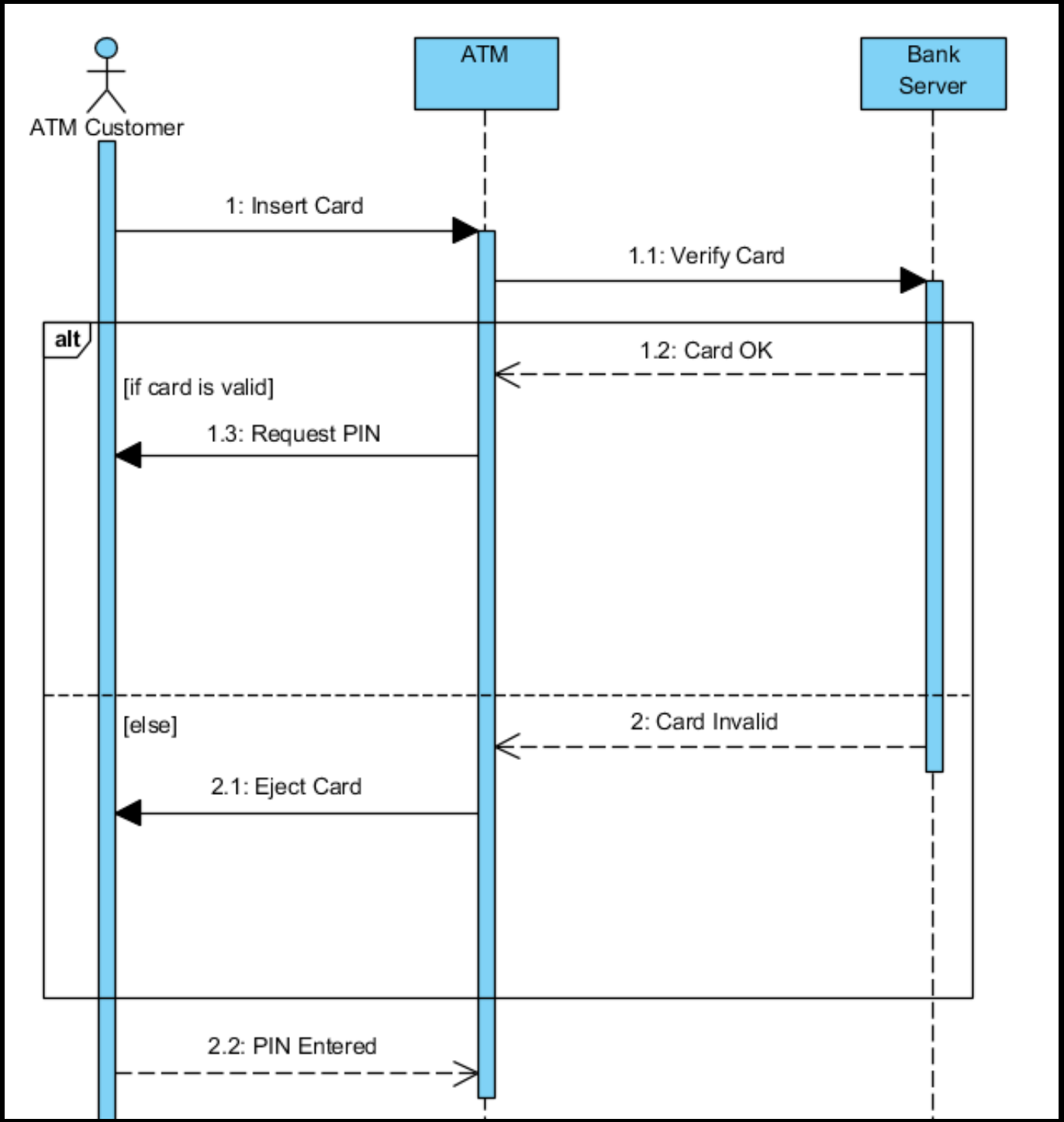
Lenguaje Unificado de Modelado

- Modelado de Casos de Uso
- Diagramas de Casos de Uso
- **Diagramas de Clases**


Person
+name: str
+phoneNumber: str
+emailAddress: str
+purchaseParkingPass()
0..1
lives at
1
Address
+street: str
+city: str
+state: str
+postalCode: int
+country: str
-validate(): bool
+outputAsLabel(): str
Student
+studentNumber: int
+averageMark: int
+isEligibleToEnroll(str): bool
+getSeminarsTaken(): int
0...*
supervises
1...5
Professor
/salary: int
#staffNumber: int
-yearsOfService: int
+numberOfClasses: int

Lenguaje Unificado de Modelado

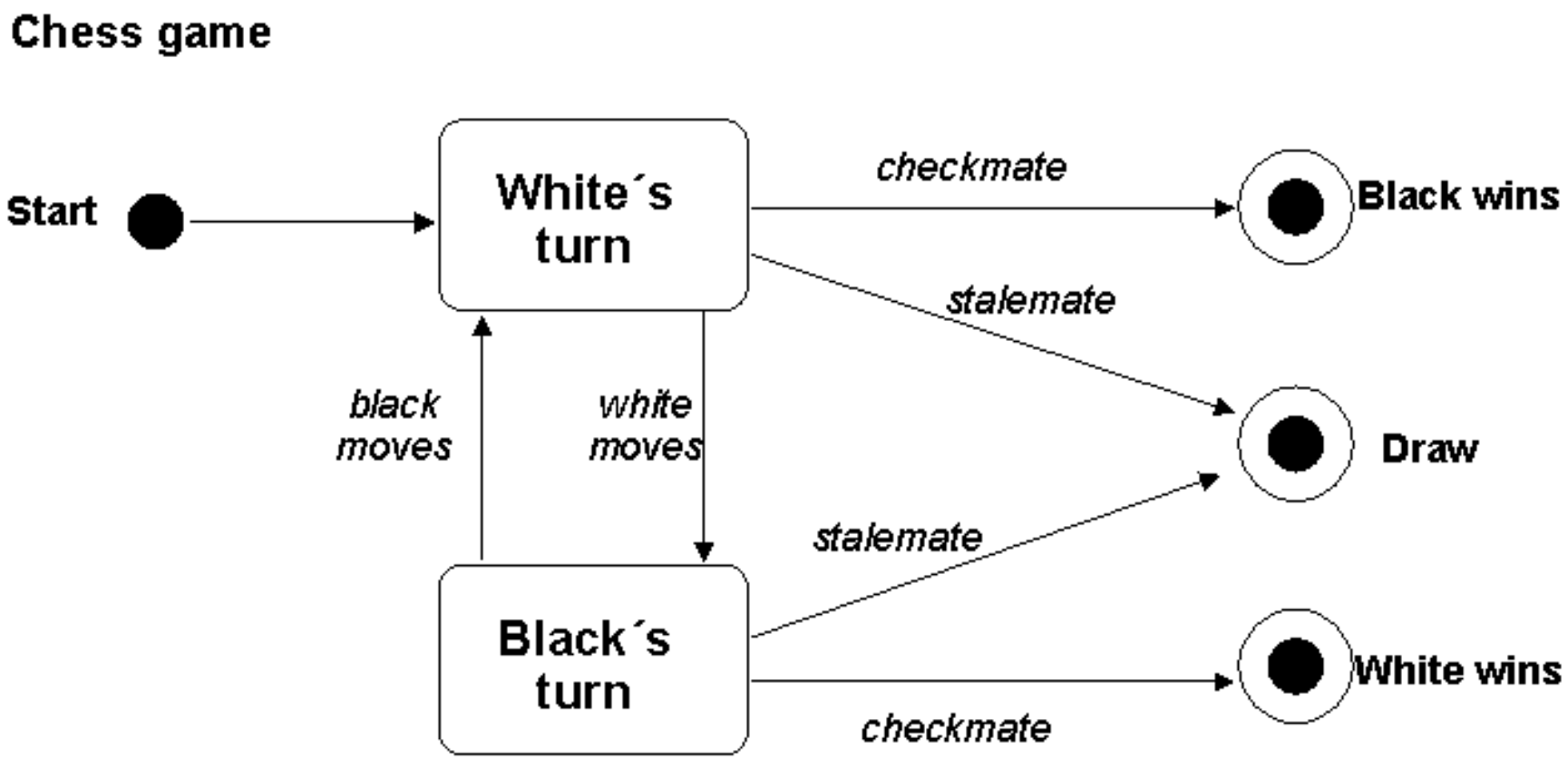
- Modelado de Casos de Uso
- Diagramas de Casos de Uso
- Diagramas de Clases
- **Diagramas de Secuencias**


ATM Customer
ATM
Bank Server
1: Insert Card
1.1: Verify Card
alt
1.2: Card OK
[if card is valid]
1.3: Request PIN
[else]
2: Card Invalid
2.1: Eject Card
2.2: PIN Entered

Lenguaje Unificado de Modelado

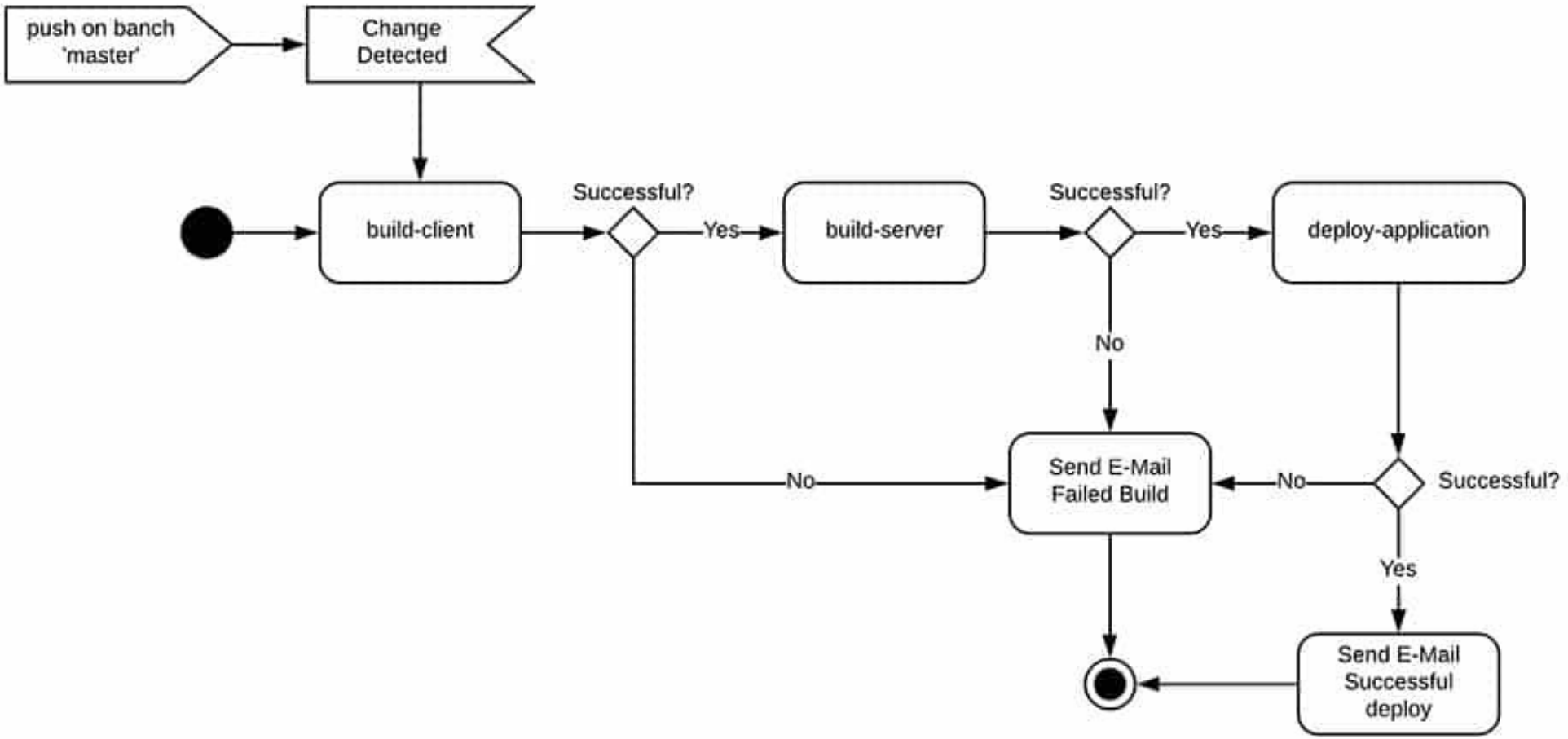
- Modelado de Casos de Uso
- Diagramas de Casos de Uso
- Diagramas de Clases
- Diagramas de Secuencias
- **Diagramas de Transición de Estado**

UML State Diagram - example



Lenguaje Unificado de Modelado

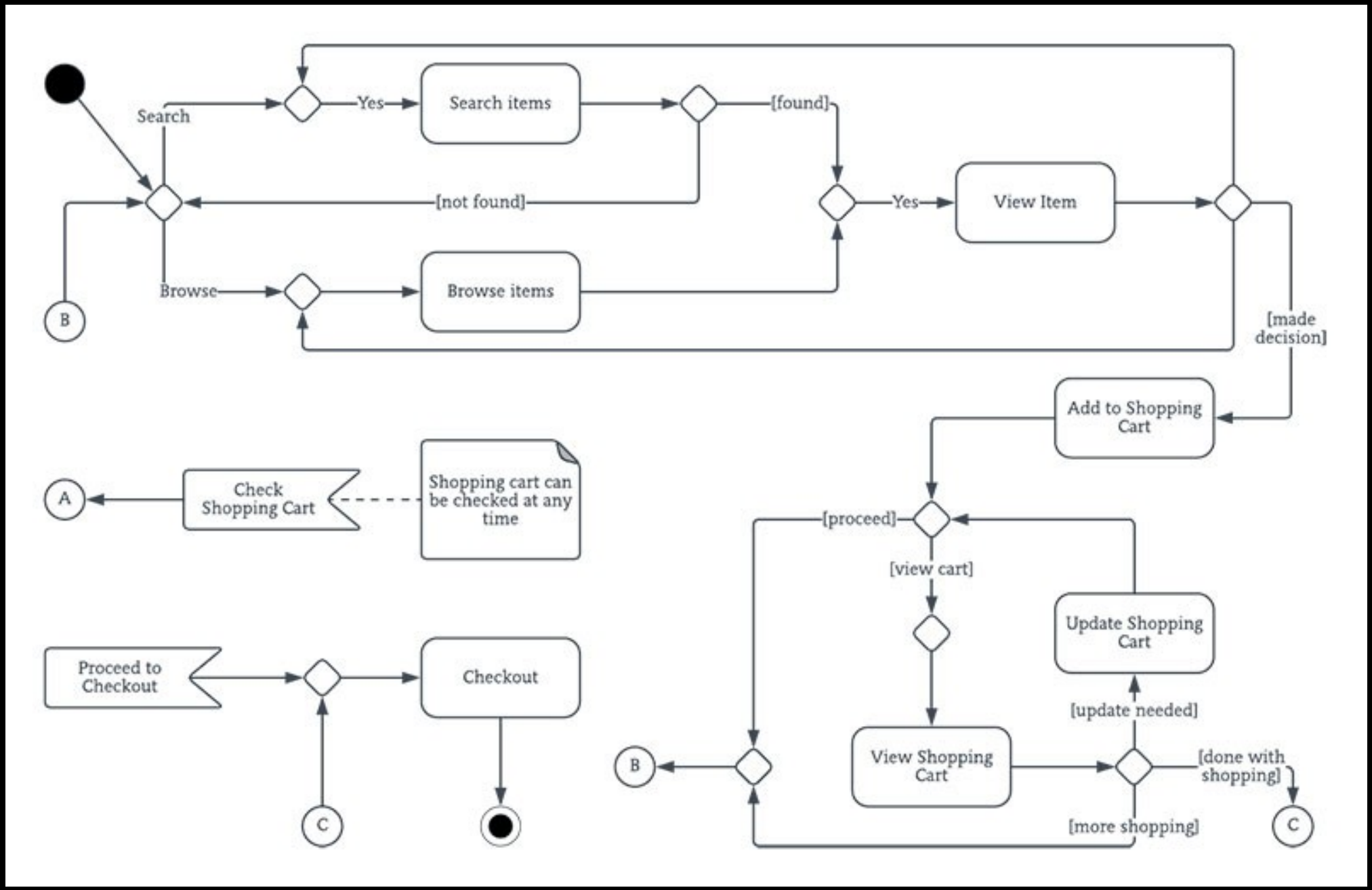
- Modelado de Casos de Uso
- Diagramas de Casos de Uso
- Diagramas de Clases
- Diagramas de Secuencias
- Diagramas de Transición de Estado
- **Diagramas de Actividades**



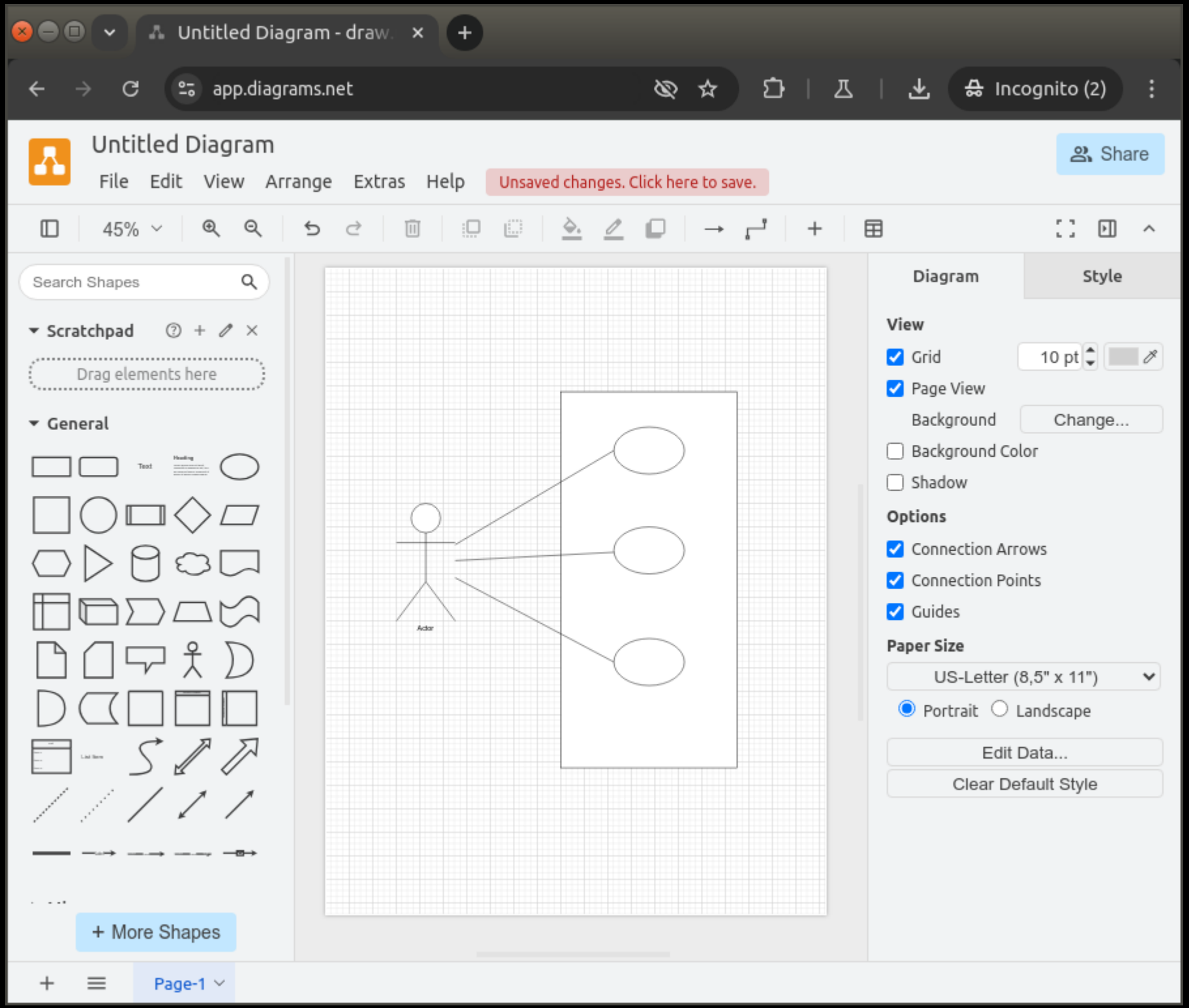
ACTIVITY DIAGRAM

Lenguaje Unificado de Modelado

- Modelado de Casos de Uso
- Diagramas de Casos de Uso
- Diagramas de Clases
- Diagramas de Secuencias
- Diagramas de Transición de Estado
- Diagramas de Actividades
- **Modelado del Proceso de Negocio**


Search
Yes
Search items
[found]
[not found]
Yes
View Item
Browse
Browse items
B
[made decision]
Add to Shopping Cart
A
Check Shopping Cart
Shopping cart can be checked at any time
[proceed]
[view cart]
Update Shopping Cart
Proceed to Checkout
Checkout
[update needed]
B
View Shopping Cart
[done with shopping]
[more shopping]
C
C

Herramientas


Untitled Diagram - draw.
app.diagrams.net
Incognito (2)
Untitled Diagram
File Edit View Arrange Extras Help
Unsaved changes. Click here to save.
Share
45%
Search Shapes
Scratchpad
Drag elements here
General
+ More Shapes
Actor
Diagram
Style
View
Grid
10 pt
Page View
Background
Change...
Background Color
Shadow
Options
Connection Arrows
Connection Points
Guides
Paper Size
US-Letter (8,5" x 11")
Portrait
Landscape
Edit Data...
Clear Default Style
Page-1